A Gallium-Fueled Fusion-Choke Applied-Field MPD Thruster: Conceptual Architecture, Performance Analysis, and Feasibility Assessment for Deep-Space Propulsion

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Abstract

Chemical propulsion imposes a fundamental mass-fraction penalty on high-Δv interplanetary missions because of the low exhaust velocities achievable through combustion. This paper presents a conceptual analysis of a hybrid deep-space propulsion architecture that combines an Applied-Field Magnetoplasmadynamic (AF-MPD) thruster with a magneto-inertial fusion (MIF) ignition stage and a capillary-fed liquid-gallium electrode system. In the concept, a magnetically confined plasma jet is compressed through a converging magnetic throat to Z-pinch conditions, briefly igniting a deuterium-tritium fusion reaction that heats a secondary propellant of liquid gallium. The gallium is delivered to refractory-metal electrodes through a porous capillary structure and is ablated by the plasma arc, cooling the electrode surface while supplying the bulk of an electromagnetically accelerated exhaust stream. This paper develops and evaluates the governing physics of the concept, including propellant mass-fraction estimates from the Tsiolkovsky rocket equation, fusion energy requirements, capillary pore sizing from the Young–Laplace relation, and neutron transport considerations for the tungsten electrode structure. The analysis indicates that exhaust velocities on the order of tens of kilometers per second could substantially reduce propellant mass fraction relative to chemical propulsion, consistent with performance ranges reported for other magneto-inertial fusion propulsion concepts in the literature. However, the assessment also identifies several unresolved engineering and programmatic constraints, including capillary pore tolerances near the limits of current additive manufacturing processes, neutron-induced transmutation of the tungsten electrode structure, exhaust-driven contamination of spacecraft surfaces, constrained global tritium supply, liquid-metal slosh dynamics, and regulatory ambiguity under existing space treaty frameworks. Realistic near-to-mid-term mission applications are more plausibly found in uncrewed cargo and science missions than in crewed transit, given the outstanding radiation, contamination, and safety-qualification questions identified. The paper concludes by outlining the ground-based manufacturing, fuel-cycle, and regulatory infrastructure required to mature the concept toward flight readiness.

Keywords: Magneto-Inertial Fusion Propulsion; Applied-Field MPD Thruster; Z-Pinch; Liquid-Metal Propellant; Capillary Porous System; Deep-Space Propulsion

Nomenclature

Δv characteristic velocity increment, m/s
v_e effective exhaust velocity, m/s
m_p propellant mass, kg
m_d dry (vehicle) mass, kg
E_k kinetic energy, J
m, v mass, kg; velocity, m/s
γ surface tension, N/m
ΔP capillary (Laplace) pressure, Pa
r pore radius, m
ρ density, kg/m³
n number density, atoms/m³ (context-dependent: gallium atomic density in Sec. 4.2; total ion density in Secs. 3.5-3.6)
σ microscopic neutron cross-section, barns (1 b = 10^{-28} m²)
λ neutron mean free path, m
N_A Avogadro constant, 6.022×10^{23} mol^{-1}
g_0 standard gravity, 9.81 m/s²

T plasma temperature, keV (or K)

V plasma volume, m^3
τ_E energy confinement (dwell) time, s
R_f fusion reaction rate density, $m^{-3}s^{-1}$
P_f fusion power, W
W_p stored plasma thermal energy, J
β plasma beta, dimensionless
B magnetic field, T
μ_0 vacuum permeability, $4\pi\times10^{-7}$ T·m/A
I pinch/coil current, A
k_B Boltzmann constant, 1.381×10^{-23} J/K
L effective current-carrying length (AF-MPD), m
η_F electromagnetic thrust-coupling factor, dimensionless

Acronyms/Abbreviations

AF-MPD – Applied-Field Magnetoplasmadynamic thruster
MIF – Magneto-Inertial Fusion
CPS – Capillary Porous System
LPBF – Laser Powder Bed Fusion
LME – Liquid Metal Embrittlement
D-T – Deuterium-Tritium
OST – Outer Space Treaty (1967)
PTBT – Partial (Limited) Test Ban Treaty (1963)
COSPAR – Committee on Space Research
ACS – Attitude Control System
NSPM-20 – National Security Presidential Memorandum 20

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1. Introduction

Interplanetary and deep-space missions are fundamentally constrained by the propellant mass fraction implied by the Tsiolkovsky rocket equation. For chemical propulsion systems, whose exhaust velocity is limited to approximately 4.4 km/s for high-performance liquid-hydrogen/liquid-oxygen combinations, even modest mission velocity increments (Δv) of several kilometers per second impose propellant mass fractions that dominate vehicle design. This constraint has motivated decades of research into higher-specific-impulse alternatives, including electric propulsion, nuclear thermal propulsion, and a range of fusion-based propulsion concepts that couple pulsed, magnetically confined fusion reactions to electromagnetic or magnetic-nozzle acceleration stages.

This paper examines a conceptual propulsion architecture that combines three previously distinct technology threads: (i) an Applied-Field Magnetoplasmadynamic (AF-MPD) thruster, which uses an externally applied magnetic field to improve thrust efficiency and electrode life relative to self-field MPD designs; (ii) a magneto-inertial fusion (MIF) ignition stage, in which a magnetically confined plasma column is compressed to fusion-relevant density and temperature through a Z-pinch process; and (iii) a capillary porous system (CPS) liquid-metal electrode, in which a refractory-metal skeleton is continuously replenished with liquid gallium delivered by capillary action. In the concept, plasma injected into a converging magnetic throat undergoes Z-pinch compression sufficient to briefly ignite a deuterium-tritium (D-T) fusion reaction; gallium ablated from the porous electrode by the resulting plasma arc absorbs the associated thermal load while supplying the bulk of the accelerated propellant mass.

The purpose of this paper is to formalize the concept's operating principles, quantify its performance envelope using standard propulsion and plasma-physics relations, and evaluate the manufacturing, materials, operational, and regulatory constraints implied by the design. Physics and engineering calculations are performed where the underlying parameters can be sourced from established reference data; where a value is assumed rather than sourced, it is explicitly flagged as a placeholder requiring further verification, consistent with a zero-fabrication standard for this manuscript. Section 2 describes the system architecture. Section 3 develops a performance analysis. Section 4 addresses manufacturing and materials constraints. Section 5 discusses operational risk factors. Section 6 addresses regulatory and planetary-protection considerations. Section 7 proposes candidate mission profiles. Section 8 outlines supporting ground infrastructure. Section 9 concludes.

2. System Architecture and Operating Principles

2.1 Overview

The proposed thruster integrates three functional subsystems into a single electromagnetic acceleration stage. First, a plasma gun injects and accelerates a deuterium-tritium fuel mixture into a magnetic throat of decreasing cross-section, where the applied magnetic field intensifies and drives geometric and magnetic compression of the plasma column. Second, the resulting Z-pinch discharge briefly ignites a fusion reaction that heats and expands the plasma; the associated thermal flux is absorbed at the electrode surface by a continuously replenished, ablating layer of liquid gallium supplied through a porous refractory-metal (tungsten or molybdenum) capillary structure. Third, a radial applied magnetic field traps the resulting energetic electron population, which electromagnetically drags the co-produced gallium ions through a magnetic nozzle to form the exhaust plume. These subsystems are described in turn below.

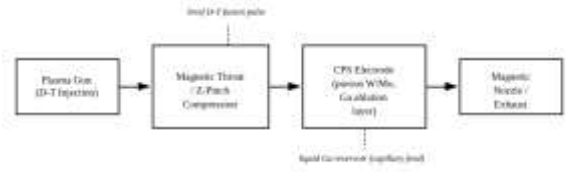


Fig. 1. System architecture: plasma injection, magnetic throat / Z-pinch compression, CPS gallium electrode, and magnetic-nozzle exhaust.

2.2 Multi-Stage Magnetic Compression and Z-Pinch Ignition

Fuel injection follows conventional plasma-gun principles: a deuterium-tritium (or deuterium–helium-3) mixture is ionized and accelerated electromagnetically to hypersonic velocity before entering the converging magnetic throat. As the plasma column advances into the region of increasing applied field strength, both the physical channel cross-section and the magnetic mirror ratio increase the confining pressure on the plasma. At the throat's narrowest point, the axial discharge current produces a self-pinching j × B force (the Z-pinch effect) that compresses the plasma column further, in principle to densities and temperatures sufficient to briefly satisfy Lawson-type ignition criteria for a D-T reaction.

The resulting fusion burn is necessarily pulsed rather than steady-state. This is consistent with the operating principle of comparable published concepts: NASA Marshall Space Flight Center's Z-pinch magneto-inertial fusion propulsion design study [1] and MSNW LLC's magneto-inertial Fusion Driven Rocket concept [2] are both organized around repetitively pulsed operation rather than continuous fusion burn. The specific pulse repetition rate assumed for this concept (20 Hz) is not independently derived or sourced in the underlying design notes.

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[Flagged placeholder: the 20 Hz pulse-repetition-rate assumption requires supporting analysis of driver electronics, thermal cycling, and structural fatigue limits before it can be treated as a validated design parameter.]

2.3 Capillary-Fed Liquid-Gallium Electrode System

A conventional solid electrode, whether tungsten, copper, or a refractory alloy, would be rapidly eroded or vaporized by direct exposure to fusion-relevant heat and particle flux. The concept instead adopts a consumable, self-replenishing electrode strategy drawn from capillary porous system (CPS) liquid-metal ion source design: the electrode is manufactured by Laser Powder Bed Fusion (LPBF) additive manufacturing as a porous refractory-metal skeleton, through which liquid gallium is continuously supplied from a propellant reservoir.

Iron-, nickel-, and chromium-based alloys, including Inconel, are unsuitable structural materials for this application because gallium is a well-documented embrittlement agent for such alloys at elevated temperature, penetrating and weakening grain boundaries through liquid metal embrittlement (LME) [3]. Tungsten and molybdenum are comparatively resistant to gallium-induced LME and are therefore the assumed skeleton materials. During operation, the plasma arc vaporizes and erodes the gallium wetting the pore surface; this ablation is deliberate rather than incidental — it removes thermal energy from the underlying refractory structure through the latent heat of vaporization while simultaneously generating the majority of the thruster's propellant mass flow. This general approach is consistent with demonstrated porous-tungsten and capillary liquid-metal ion source designs used in electric propulsion and ion-beam applications [4].

2.4 Plasma Acceleration and Charge Configuration

Two charge-configuration strategies were considered in the underlying concept notes. Biasing the outer shell strongly positive to repel gallium ions and preferentially accelerate electrons was rejected because it is subject to the space-charge limit: a dense electron sheath forms at the wall, both shielding the applied field and exposing the structure to destructive electron bombardment. The adopted configuration instead follows conventional Hall- and MPD-thruster practice [5], in which a radial applied magnetic field traps energetic electrons in an azimuthal drift layer. This electron population both sustains ionization of the gallium sweating from the electrode surface and establishes the axial electric field that electromagnetically accelerates the resulting gallium ions rearward through a magnetic nozzle, producing a collimated exhaust plume.

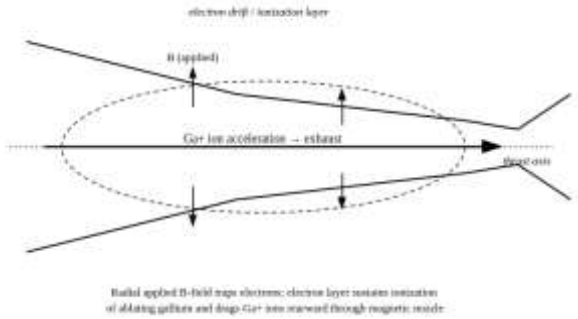


Fig. 2. Radial applied-field charge configuration: electron drift layer sustains ionization and drags Ga+ ions through the magnetic nozzle.

3. Performance Analysis

3.1 Exhaust Velocity Regime and Comparison to Published Concepts

The concept notes propose an electromagnetic exhaust velocity range of 20–100 km/s, driven by the combination of heavy gallium ions (atomic mass ≈ 69.7 u) with fusion-augmented acceleration. This range corresponds to a specific impulse (Isp = ve/g0) of approximately 2,040–10,190 s. For comparison, NASA Marshall Space Flight Center's Z-pinch magneto-inertial fusion propulsion concept design study reports an Isp near 19,436 s at a thrust level of several kN [1], while MSNW LLC's magneto-inertial Fusion Driven Rocket concept reports an Isp range of approximately 1,606–5,722 s depending on assumed fusion gain, with burn durations of one to ninety days [2].

The gallium-fusion concept's assumed performance range therefore falls within, or close to, the range reported for comparable published magneto-inertial fusion propulsion studies, which provides qualitative plausibility for the assumed exhaust velocity but does not by itself validate the specific engine design proposed here. A first-principles derivation of achievable exhaust velocity from the concept's magnetic field topology, pinch current, and gallium ionization fraction remains outstanding and is recommended as follow-on work. For the calculations in Sections 3.2–3.3, a representative exhaust velocity of ve = 50 km/s (Isp ≈ 5,097 s) is adopted as a mid-range design point, consistent with the value used in the underlying concept notes.

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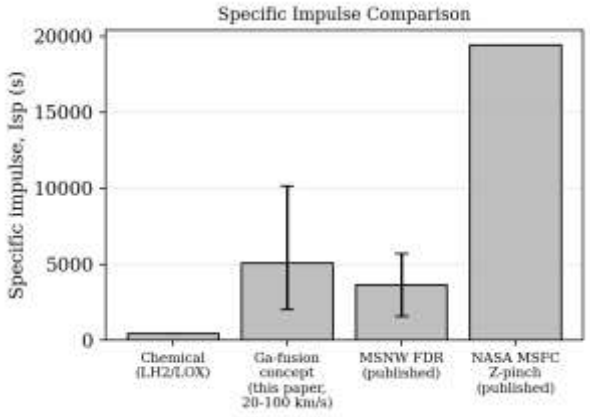


Fig. 3. Specific impulse of this concept (range) versus chemical propulsion and published MIF concepts (MSNW FDR; NASA MSFC Z-pinch).

3.2 Propellant Mass-Fraction Estimate

The Tsiolkovsky rocket equation relates the propellant mass mp required to achieve a velocity increment Δv, given dry vehicle mass md and effective exhaust velocity ve:

$$mp = md \times (e^{(\Delta v / ve)} - 1) \quad (1)$$

For an illustrative dry mass of md = 100 t and a representative impulsive trans-Mars-injection Δv of 8 km/s, a chemical propulsion system with ve = 4.4 km/s requires an exponent Δv/ve = 1.82, giving mp ≈ 5.16 md ≈ 516 t. Under the same assumptions, the gallium-fusion concept's ve = 50 km/s design point gives an exponent of 0.16 and mp ≈ 0.17 md ≈ 17 t — an approximately 30-fold reduction in propellant mass for the same mission Δv. Both figures were independently recomputed for this paper and are consistent with the underlying concept notes.

This comparison carries an important caveat that is not addressed in the underlying concept notes. The Tsiolkovsky relation in its impulsive form is appropriate for short, high-thrust burns, as is characteristic of chemical propulsion. Magneto-inertial fusion propulsion concepts of the type considered here instead produce thrust at milli-g-class accelerations; NASA Marshall Space Flight Center's Z-pinch propulsion trajectory analysis explicitly treats this regime as “medium thrust” and requires numerically integrated, patched-conic trajectory analysis rather than an impulsive Δv budget [1]. A low-thrust vehicle typically requires a larger effective Δv than an equivalent impulsive trajectory, because of gravity losses accrued during a slow escape spiral out of a planetary gravity well. The 8 km/s, 17 t propellant estimate above should therefore be read as an optimistic lower bound; a credible mission-level propellant budget for this concept requires low-thrust trajectory optimization rather than the impulsive rocket equation alone.

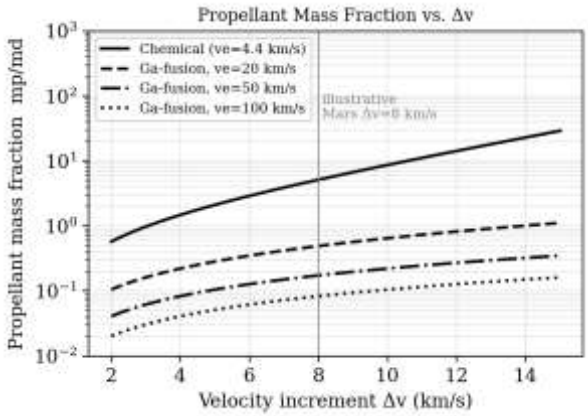


Fig. 4. Propellant mass fraction vs. Δv (Tsiolkovsky equation) for chemical propulsion and this concept's exhaust-velocity range.

3.3 Fusion Fuel Mass per Pulse

An independent scenario in the concept notes considers the D-T fuel mass required to accelerate a 65 t gallium propellant load to the design exhaust velocity of 50 km/s. The kinetic energy required is:

$$Ek = \tfrac{1}{2} m v^2 = \tfrac{1}{2}(65{,}000\ \text{kg})(50{,}000\ \text{m/s})^2 \approx 8.1\times10^{13}\ \text{J} \quad (2)$$

D-T fusion releases 17.6 MeV per reaction. Given the combined mass of one deuteron and one triton (≈ 5.03 u ≈ 8.35×10^{-27} kg), this corresponds to a fusion energy density of approximately 3.4×10^{14} J per kilogram of D-T fuel consumed, a figure independently recomputed for this paper from standard nuclear reaction data and consistent with the concept notes. Dividing the required kinetic energy by this energy density gives a nominal D-T fuel mass of:

$$mDT = Ek / (3.4\times10^{14}\ \text{J/kg}) \approx 0.24\ \text{kg} \quad (3)$$

This calculation, however, implicitly assumes that 100% of the fusion yield is converted into propellant kinetic energy, with no losses to radiation, imperfect magnetic confinement, incomplete burn fraction, or inefficient momentum coupling between the fusion plasma and the gallium propellant. Conceptual Z-pinch propulsion studies in the literature explicitly model these coupling losses and report substantially lower net propulsive efficiencies than an idealized 100% conversion would imply [6].

[Flagged placeholder: the fusion-to-propellant energy coupling efficiency assumed for this concept is not stated in the underlying design notes — Insert Data Here. The 0.24 kg D-T fuel-mass figure should accordingly be treated as a strict lower bound rather than an engineering estimate.]

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[Flagged conflict: the 65 t gallium propellant load used in this scenario is not reconciled with the 17 t propellant-mass-fraction result obtained in Section 3.2, which was derived under a separate set of assumptions (100 t dry mass, 8 km/s impulsive Δv). These two figures originate from independent illustrative scenarios in the underlying concept notes rather than a single, self-consistent vehicle mass budget, and should be reconciled before being combined into an overall vehicle sizing estimate.]

3.4 Staged Energy and Power Balance

Section 3.3 computed the D-T fuel mass required to accelerate the 65 t gallium propellant load to ve = 50 km/s under an idealized 100% fusion-to-propellant energy conversion. This section replaces that idealization with a staged energy budget and an explicit, labeled coupling-efficiency parameter, per the zero-fabrication standard used throughout this manuscript.

Define the overall fusion-to-propellant coupling efficiency:

$$\eta_total = E_useful / E_fusion \quad (6)$$

where E_useful is the useful exhaust kinetic energy delivered to the propellant (taken as Ek from Eq. 2, 8.1×10^{13} J, for the same 65 t / 50 km/s baseline scenario used in Section 3.3) and E_fusion is the total D-T fusion energy that must be released to deliver it. Between fusion energy release and useful exhaust kinetic energy, energy is diverted through several loss channels: (i) gallium heating, vaporization, and ionization at the ablating electrode surface; (ii) incomplete D-T burn fraction (quantified directly in Section 3.5); (iii) the 14.1 MeV neutron carrying 80.1% of the 17.6 MeV D-T yield, most of which escapes the thin gallium ablation layer without depositing energy, consistent with the ~9-10 cm neutron mean free path estimated in Section 4.2; (iv) radiation (bremsstrahlung, line radiation) and conductive thermal losses from the pinch plasma; (v) resistive/Ohmic and momentum-coupling losses in the electromagnetic acceleration stage (Section 2.4); and (vi) magnetic-nozzle divergence and incomplete plasma detachment.

Independent, validated values for each of these six channels are not established in the underlying concept notes or in the literature cited for this specific engine architecture. Consistent with the zero-fabrication standard, this paper does not multiply out a bottom-up product of individually fabricated sub-efficiencies; instead, η_total is treated directly as a swept design parameter over η_total = 5%, 10%, 20%, 30%, with 10% adopted as the nominal engineering assumption. This nominal value is broadly consistent with the substantially-lower-than-ideal net propulsive efficiencies reported for comparable pulsed Z-pinch propulsion coupling analyses in the literature [6], but is not independently derived for this architecture.

[Flagged placeholder: per-channel efficiency values (heating/vaporization/ionization loss, burn fraction, neutron loss, radiation loss, EM coupling loss, nozzle loss) are not independently quantified in this paper; only their aggregate product, η_total, is treated as an assumed sensitivity-swept parameter — Insert Data Here: per-channel coupling-efficiency breakdown from dedicated simulation.]

Table 1 also requires a total-mission burn duration to convert the total energy/fuel-mass figures below into average fusion power and per-pulse energy at the 20 Hz repetition rate assumed in Section 2.2. This duration is not stated in the underlying concept notes; a 1-90 day bracket is adopted here by analogy to the burn-duration range reported for MSNW's comparable magneto-inertial Fusion Driven Rocket concept (Section 3.1) [2], rather than as a value derived from this concept's own design notes.

Table 1. Energy and power balance vs. η_total (baseline: 65 t Ga, ve = 50 km/s, Eq. 2 gives Euseful = Ek = 8.125×10^{13} J; 20 Hz pulse rate; burn-duration bracket 1-90 days [2]; nominal case in bold).

η_total	E_fusion (J)	m_DT (kg)	P_fusion,avg (W)
5%	1.63×10^{15}	4.78	1.9×10^{10}-2.1×10^{8}
10% (nom.)	8.13×10^{14}	2.39	9.4×10^{9}-1.1×10^{8}
20%	4.06×10^{14}	1.19	4.7×10^{9}-5.2×10^{7}
30%	2.71×10^{14}	0.80	3.1×10^{9}-3.5×10^{7}

P_fusion,avg spans the 1-90 day burn-duration bracket; the corresponding energy per pulse at 20 Hz is P_fusion,avg / 20 in each case (e.g., nominal case: 4.70×10^{8} J/pulse at the 1-day bound, 5.22×10^{6} J/pulse at the 90-day bound). The useful (post-efficiency) exhaust power is, by definition, independent of η_total: Puseful,avg = Euseful / t_burn = 9.40×10^{8} W (1-day burn) to 1.04×10^{7} W (90-day burn), with a corresponding useful energy per pulse of 4.70×10^{7} J to 5.22×10^{5} J. The required fusion power and D-T fuel mass instead scale as 1/η_total. The nominal (η_total = 10%) fusion-power bracket, 1.05×10^{8} - 9.40×10^{9} W, is carried forward as the required fusion power for the ignition check in Section 3.5.

3.5 Fusion Ignition and Lawson Criterion Analysis

This section tests whether a D-T plasma consistent with the pulsed Z-pinch compression described in Section 2.2 could plausibly produce the required fusion power identified in Section 3.4, without assuming ignition. The underlying concept notes do not specify plasma temperature, density, volume, or confinement time; the following values are adopted as order-of-magnitude assumptions by analogy to the Z-pinch and magneto-inertial fusion (MIF) concepts already cited in this paper [1,2,6], and are flagged accordingly rather than treated as validated design values: plasma temperature T ≈ 10 keV;

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total ion (deuteron + triton) density $n \approx 1\times10^{25}$ m^{-3}, with an approximately equal (50/50) D/T mixture so that $n_D = n_T = n/2$; a compressed pinch-column volume $V \approx 4\times10^{-6}$ m^3 (illustrative cylindrical geometry, radius ≈ 0.5 cm, length ≈ 5 cm); and a confinement (compression dwell) time $\tau_E \approx 1$ μs, representative of pulsed Z-pinch/MIF stagnation timescales rather than steady-state magnetic confinement.

[Flagged placeholder: T, n, V, and τ_E are order-of-magnitude assumptions by analogy to cited MIF/Z-pinch literature, not values derived from this concept's own design notes or from a device-specific simulation — Insert Data Here: verified plasma parameters from dedicated Z-pinch compression modeling.]

For an approximately equal D/T mixture, the fusion reaction rate density is:

$$Rf = nD\cdot nT\cdot\langle\sigma v\rangle = (n^2/4)\langle\sigma v\rangle (7)$$

Using the standard D-T reactivity $\langle\sigma v\rangle \approx 1.1\times10^{-22}$ m^3/s at T = 10 keV: $Rf = (1\times10^{25})^2/4 \times 1.1\times10^{-22} \approx 2.75\times10^{27}$ reactions$\cdot m^{-3}\cdot s^{-1}$. The corresponding fusion power is:

$$Pf = Rf\cdot V\cdot EDT \approx (2.75\times10^{27})(4\times10^{-6}\ m^3)(2.82\times10^{-12}\ J) \approx 3.05\times10^{10}\ W\ (30.5\ GW) (8)$$

This instantaneous fusion power (30.5 GW) is of the same order of magnitude as, and exceeds, the nominal required fusion power bracket from Section 3.4 (1.05×10^8 - 9.40×10^9 W), indicating no fundamental power-magnitude mismatch at this assumed n, T, V operating point. This comparison alone, however, only tests whether the assumed reacting state — if achieved — could supply enough power; it does not test whether that state is itself achievable, which the Lawson analysis below addresses directly.

The Lawson triple product for the assumed state is:

$$n\cdot T\cdot\tau E = (1\times10^{25}\ m^{-3})(10\ keV)(1\times10^{-6}\ s) \approx 1\times10^{20}\ keV\cdot s\cdot m^{-3} (9)$$

The standard D-T ignition-relevant triple-product criterion is commonly cited in the range $nT\tau E \gtrsim 3\times10^{21}$ $keV\cdot s\cdot m^{-3}$ (order-of-magnitude; reported values vary by convention and required gain), roughly 30× above the value calculated here. The simpler Lawson breakeven criterion ($n\tau E \gtrsim \sim1.5\times10^{20}$ $s\cdot m^{-3}$ near optimal temperature) is likewise not met: $n\tau E = (1\times10^{25})(1\times10^{-6}) = 1\times10^{19}$ $s\cdot m^{-3}$, about an order of magnitude below breakeven. Under these assumptions, the plasma state does NOT satisfy either the Lawson breakeven or the ignition triple-product criterion; this result is reported directly rather than adjusted to force a positive outcome.

The approximate stored plasma thermal energy is:

$$Wp \approx 3nkBT\cdot V \approx 3(1\times10^{25})(1.60\times10^{-15}\ J)(4\times10^{-6}\ m^3) \approx 1.89\times10^5\ J (10)$$

Relating this to confinement/loss power through $\tau E = Wp/Ploss$ and inverting for the assumed $\tau E \approx 1$ μs gives an implied loss power $Ploss \approx Wp/\tau E \approx 1.89\times10^{11}$ W (189 GW) — roughly six times larger than the fusion power Pf calculated above. This is internally consistent with the sub-ignition finding: a plasma approaching ignition would have alpha-heating power comparable to or exceeding confinement losses, whereas here total fusion power (let alone the smaller alpha-only fraction of it) is well below the power needed to sustain the assumed state for τE, confirming this operating point is confinement-loss-dominated rather than self-heating-dominated.

Of the 17.6 MeV D-T yield, the 3.5 MeV alpha particle carries 19.9% (≈6.06 GW of Pf) and the 14.1 MeV neutron carries 80.1% (≈24.4 GW of Pf). Consistent with Section 4.2's neutron mean-free-path estimate, most neutron energy is expected to escape the compressed plasma and thin gallium layer rather than depositing locally. The fraction of alpha energy actually deposited within the assumed centimeter-scale, microsecond-duration pinch volume is not established in this paper or in the literature reviewed.

[Flagged placeholder: alpha energy deposition fraction within the assumed plasma volume and confinement time requires simulation/experimental validation and is not estimated here.]

Distinguishing breakeven, ignition, and engineering feasibility: (1) Breakeven/ignition, in the strict Lawson sense of a self-sustaining, alpha-heating-dominated plasma, is NOT satisfied by the assumed parameters — both the triple product and the simpler nτE criterion fall short by roughly one to two orders of magnitude, and the implied confinement-loss power exceeds the fusion power produced. (2) This does not by itself disqualify the concept as described, because Section 2.2 already characterizes the design as an externally driven, pulsed Z-pinch rather than a self-igniting steady-state device: the pulsed compression driver, not alpha self-heating, is what establishes the assumed n, T state each cycle, and the D-T yield is properly understood as a supplementary, not dominant, energy contribution relative to that external driver. (3) Engineering feasibility remains undetermined: whether the assumed n, T, V, τE operating point is itself physically achievable by the described magnetic-throat/Z-pinch compression mechanism, and at what driver energy cost, is not established by this paper and requires dedicated compression-dynamics simulation.

3.6 Plasma Confinement and Magnetic Pressure Requirements

Using the plasma n and T assumed in Section 3.5, and adopting the total (electron + ion) particle density convention for a fully ionized, quasineutral D-T plasma

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(ne = ni = n for singly-charged D and T ions, so total particle density = 2n), the plasma pressure is:

$$P_{plasma} = (n_e + n_i)kBT = 2n \cdot kBT \approx 2(1\times10^{25})(1.60\times10^{-15}\ J) \approx 3.20\times10^{10}\ Pa\ (32.0\ GPa) \quad (11)$$

The confining magnetic field required to balance this pressure at a given plasma beta $\beta = P_{plasma}/P_B$ is:

$$B_{required} = \sqrt{(2\mu_0 P_{plasma} / \beta)} \quad (12)$$

Adopting $\beta \approx 0.5$ as a representative high-beta operating point (flagged assumption, consistent with the high-beta Z-pinch/FRC-like operation typical of the MIF concepts cited in this paper [1,2,6]) gives $B_{required} \approx 401$ T. The underlying concept notes describe an "applied magnetic field" qualitatively (Sections 2.2-2.4) but do not state a numeric design field, so this figure cannot be checked against a stated value and is flagged as requiring simulation/experimental validation against an actual coil/pinch-driver design. Quantitatively, ~400 T is far beyond the ~10-20 T practical ceiling for continuously energized superconducting magnets, but is of the same order as pulsed, self-generated Z-pinch fields at the multi-megaampere current level computed below — consistent with the pulsed, rather than continuously applied, nature of the compression field already described in Section 2.2.

[Flagged placeholder: $\beta \approx 0.5$ is an assumed operating point, not a value stated in the underlying concept notes or independently derived for this architecture — Insert Data Here: verified plasma beta from device-specific simulation.]

The corresponding magnetic energy stored in the pinch volume is:

$$U_B = (B^2/2\mu_0)\cdot V \approx 2.52\times10^5\ J\ (252\ kJ) \quad (13)$$

For the pinch-column (Z-pinch) geometry already described in Section 2.2, the appropriate field relation is the cylindrical-current Ampere's-law result for the azimuthal self-field of an axial pinch current — not a solenoid or open straight-wire-in-vacuum model — evaluated at the pinch radius $r \approx 0.5$ cm assumed in Section 3.5:

$$B_{required} = \mu_0 I / (2\pi r) \implies I \approx 1.00\times10^7\ A\ (10.0\ MA) \quad (14)$$

A required pinch current of order 10 MA is broadly consistent, in order of magnitude, with the multi-to-tens-of-megaampere currents used in existing pulsed-power Z-pinch research facilities, which supports the plausibility of the current magnitude in isolation. Sustaining a discharge of this magnitude repetitively at the 20 Hz rate assumed in Section 2.2 (rather than as a single-shot pulse, as in existing Z-pinch facilities) is a substantially different and currently unresolved pulsed-power engineering regime.

The associated magnetic-pressure structural stress on any confining structure is:

$$\sigma_B \sim B^2/2\mu_0 = P_{plasma}/\beta \approx 6.41\times10^{10}\ Pa\ (64.1\ GPa) \quad (15)$$

This is roughly 60-200× the ultimate tensile strength of tungsten or molybdenum alloys (typically a few hundred MPa to ~1 GPa), the candidate electrode structural materials identified in Section 2.3/4.1. A static structure could not continuously withstand this magnetic pressure. Because the concept is inherently pulsed rather than steady-state — in the manner of existing Z-pinch experiments, where confinement during the brief current pulse is provided by plasma/liner inertia rather than a static containment vessel — the governing engineering question is dynamic/fatigue loading over a sub-microsecond-to-microsecond pulse rather than static stress. This reframing does not resolve the constraint; it remains unquantified and is flagged as requiring dedicated pulsed-structural analysis.

[Flagged placeholder: dynamic/fatigue structural response of the electrode and coil structure under repetitive ~64 GPa-equivalent pulsed magnetic loading at 20 Hz requires simulation/experimental validation and is not assessed in this paper.]

Master Parameters (Sections 3.4-3.6)

The following table consolidates the parameters used across the energy-balance, Lawson, and magnetic-confinement analyses above, per the consistency rules for this revision. Where the paper's underlying notes contain two independently sourced values for the same quantity (the 65 t and 17 t propellant-mass scenarios), both are retained and their distinct purposes are labeled explicitly rather than one being arbitrarily discarded, consistent with the flagged conflict already noted in Section 3.3.

Parameter (basis/status)	**Value**
Propellant mass, energy-budget scenario (Sec. 3.3 baseline)	m_{Ga} = 65,000 kg
Propellant mass, Δv/mass-fraction scenario (Sec. 3.2 baseline; independent, not reconciled with m_{Ga})	$m_p \approx$ 17,000 kg
Exhaust velocity (Sec. 3.1 design point)	v_e = 50 km/s
Pulse repetition rate (Sec. 2.2, flagged, unvalidated)	f = 20 Hz
Total-mission burn duration (assumed, by analogy to ref. [2])	1-90 days
Coupling efficiency (assumed, swept, Sec. 3.4)	η_{total} = 5/10/20/30%, nom. 10%
D-T fusion yield (standard data)	E_{DT} = 17.6 MeV
Plasma temperature (assumed, by analogy [1,2,6])	T ≈ 10 keV
Total ion density D+T (assumed, by analogy [1,2,6])	$n \approx 1\times10^{25}$ m^{-3}
D/T mixture (assumed equal mix)	≈50/50 by number
Pinch-column volume (illustrative geometry assumption)	$V \approx 4\times10^{-6}$ m³ ($r \approx 0.5$cm, $L \approx 5$cm)
Confinement/dwell time (assumed, by analogy [1,2])	$\tau_E \approx 1$ μs
Plasma beta (assumed, high-beta pinch/FRC-like [1,2,6])	$\beta \approx 0.5$
Required confining field (calculated, Sec. 3.6; no paper-stated field to compare against)	$B_{required} \approx$ 401 T
Required pinch current (calculated, Sec. 3.6, Z-pinch geometry)	$I_{required} \approx$ 10.0 MA

3.7 Physics Feasibility Summary

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1. Required fusion power (nominal ηtotal = 10%): ≈1.05×10^{8} - 9.40×10^{9} W average, depending on the assumed 1-90 day burn duration bracket, with ≈5.2×10^{6} - 4.7×10^{8} J per pulse at 20 Hz.

2. Assumed plasma state: T ≈ 10 keV, n ≈ 1×10^{25} m^{-3}, τE ≈ 1 μs — order-of-magnitude assumptions by analogy to cited MIF/Z-pinch literature, not independently derived or simulated for this architecture.

3. Lawson result: nTτE ≈ 1×10^{20} keV·s·m^{-3}, roughly 30× below the ~3×10^{21} keV·s·m^{-3} ignition-relevant triple product; the simpler nτE criterion is also not met. Ignition/breakeven is NOT satisfied under these assumptions; the implied confinement-loss power (≈189 GW) exceeds the calculated fusion power (≈30.5 GW).

4. Required magnetic field: Brequired ≈ 401 T (β ≈ 0.5 assumption) with Irequired ≈ 10.0 MA pinch current; the associated magnetic-pressure structural stress (≈64.1 GPa) is far beyond the static strength of candidate electrode materials, though partially reframed (not resolved) by the concept's inherently pulsed operation.

5. Overall assessment: marginal/uncertain, trending toward inconsistent with the calculated requirements as currently specified. The assumed plasma state does not reach ignition; the required field and structural stress far exceed continuous/static engineering limits; and while the instantaneous fusion power implied by the assumed reacting state is of a plausible order of magnitude relative to the mission energy requirement, this does not establish that the assumed n, T, τE state is achievable by the Z-pinch compression mechanism as described. None of these findings indicate a fundamental physical impossibility, but they substantially raise the bar for the simulation and experimental validation work already recommended in Section 9, and should be read alongside it rather than as a separate conclusion.

3.8 AF-MPD Thrust Derivation and Consistency Check

This section derives the electromagnetic thrust produced by the AF-MPD acceleration stage described in Sections 2.4 and 3.1, and checks whether the resulting exhaust velocity and power are internally consistent with the ve = 50 km/s design point used elsewhere in this paper. This check is independent of, and does not modify, the energy-balance (3.4), Lawson (3.5), or magnetic-confinement (3.6) analyses; it reuses their numerical outputs only as inputs, as noted explicitly below.

3.8.1 Force model

The electromagnetic body force on the current-carrying plasma is, exactly:

$$F = \int (J \times B)\, dV \quad (16)$$

For a first-order engineering estimate, the plasma column is approximated as carrying a uniform discharge current I over an effective current-carrying length L, threading a uniform applied field of magnitude B at angle θ to the current direction. This is NOT an exact description of the plasma (which has nonuniform J and B, finite divergence, and complex current paths); it is a lumped-parameter approximation standard in MPD-thruster engineering analysis [5]:

$$F \approx I\, L\, B \sin(\theta) \quad (17)$$

where I is the discharge/plasma current (A), L is the effective current-carrying length (m), B is the applied magnetic-field magnitude (T), and θ is the angle between current and field (θ = 90° for the perpendicular, force-maximizing geometry assumed here as an engineering default).

3.8.2 Electromagnetic coupling factor

Real plasma current density and magnetic-field distributions are nonuniform, the plasma diverges rather than forming an ideal column, and some current paths (e.g., near-electrode return currents) do not contribute directly to axial thrust. These effects are captured with an effective electromagnetic thrust-coupling factor ηF, defined as the ratio of the actual axial electromagnetic force to the idealized F = ILB sin(θ) estimate:

$$F_{actual} = \eta_F \cdot I\, L\, B \sin(\theta) \quad (18)$$

ηF is not a generic "loss factor" but a specific measure of how effectively the applied current and field geometry converts into useful axial thrust, encompassing current-density nonuniformity, field nonuniformity, plasma divergence half-angle, and non-axial current paths. Neither this paper's underlying concept notes nor the AF-MPD reference cited elsewhere in this paper [5] provide a validated ηF for this specific geometry, so no definitive value is assigned. Section 3.8.3 instead solves for the ηF required to reproduce the paper's existing performance assumptions, and Section 3.8.4 evaluates that required value for physical plausibility, rather than assuming a preliminary range (e.g., 0.5-1.0) without justification.

3.8.3 Exhaust velocity and thrust: predicted vs. existing

The electromagnetic force couples to the exhaust stream through momentum flux:

$$F = \dot{m} \cdot v_e \implies v_e = F/\dot{m} \quad (19)$$

A mass flow rate is not independently stated in the underlying concept notes. Consistent with the burn-duration bracket already adopted in Section 3.4 [2], $\dot{m}$ = mGa/t_burn using the 65 t baseline gives $\dot{m}$ ≈ 0.752 kg/s (1-day burn) to ≈8.36×10^{-3} kg/s (90-day burn).

The only I and B values established anywhere in this paper are the Z-pinch confinement values from Section 3.6 (I ≈ 10.0 MA, B ≈ 401 T), which were derived for the fusion-ignition pinch geometry, not independently for the AF-MPD acceleration stage of Section 2.4. No separate AF-

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MPD-stage current or field is given. Using these values with the illustrative pinch length L ≈ 0.05 m (Section 3.5/3.6 geometry assumption) as the only available proxy for an effective conductor length, and θ = 90°, the idealized (ηF = 1) force is:

$$F_{ideal} = I\,L\,B \approx (1.00\times10^{7}\ A)(0.05\ m)(401\ T) \approx 2.01\times10^{8}\ N \quad (20)$$

This is an upper bound (ηF ≤ 1). Dividing by the mass flow rates above gives the AF-MPD-model-predicted exhaust velocity:

$$v_{e,predicted} = F_{ideal} / \dot{m} \approx 2.67\times10^{8}\ m/s\ (89\%\ of\ c),\ 1\text{-day burn};\ \approx 2.40\times10^{10}\ m/s\ (>80\times\ c),\ 90\text{-day burn} \quad (21)$$

Both results are unphysical (the 90-day case exceeds the speed of light by more than an order of magnitude; the 1-day case is a substantial fraction of c, invalidating the non-relativistic momentum treatment used here). This is reported directly, per the zero-fabrication standard, rather than adjusted. It indicates that the Z-pinch confinement I and B (Section 3.6) cannot be the actual continuous, or pulse-averaged, electromagnetic acceleration parameters acting on the gallium exhaust — those confinement values describe a brief (~1 μs, Section 3.5), extreme-magnitude pulse needed momentarily for fusion ignition, not a sustained or repetitively-averaged accelerating force on a much larger, slower mass flow. Treating them as interchangeable is not physically justified and is the direct cause of the inconsistency found here.

Conversely, solving Eq. 18-19 for the ηF required to reproduce the existing ve = 50 km/s design point, given $F_{existing} = \dot{m}\cdot v_e$:

$$F_{existing} = \dot{m}\cdot v_e \approx 3.76\times10^{4}\ N\ (1\text{-day}),\ \approx 4.18\times10^{2}\ N\ (90\text{-day}) \quad (22)$$

$$\eta F,required = F_{existing} / F_{ideal} \approx 1.87\times10^{-4}\ (1\text{-day}),\ \approx 2.08\times10^{-6}\ (90\text{-day}) \quad (23)$$

These required ηF values (on the order of 10^{-4}-10^{-6}) are far below any physically reasonable range for an electromagnetic coupling factor, which by construction cannot exceed 1 and would not plausibly fall below roughly 0.1-0.5 for a functioning MPD-class thruster [5]. An implausibly small ηF is not a valid resolution; it is a symptom confirming the same conclusion as Eq. 21: the Section 3.6 pinch-confinement I and B are not an appropriate stand-in for the AF-MPD acceleration-stage parameters, and this paper does not currently establish the latter independently.

[Flagged inconsistency: the AF-MPD thrust predicted from the only I, B values available in this paper (Section 3.6, Z-pinch confinement stage) is inconsistent with the existing ve = 50 km/s / mass-flow assumptions (Sections 3.1, 3.4) by 4-6 orders of magnitude, and the exhaust-velocity prediction is non-physical (exceeds c). This does not necessarily invalidate the ve = 50 km/s design point itself (which is independently supported by comparison to published MIF concepts, Section 3.1), but it does mean the AF-MPD electromagnetic force has not been validated against that design point from first principles. Insert Data Here: independently specified AF-MPD acceleration-stage current, field, and effective length, distinct from the Z-pinch confinement parameters, from dedicated circuit/field design or simulation.]

3.8.4 Energy consistency check

Using the existing ve = 50 km/s and the mass flow ṁ derived above (not the non-physical predicted ve from Eq. 21), the kinetic power is:

$$P_{kinetic} = \tfrac{1}{2}\dot{m}v_e^{2} = \tfrac{1}{2}F\cdot v_e \approx 9.40\times10^{8}\ W\ (1\text{-day}),\ \approx 1.04\times10^{7}\ W\ (90\text{-day}) \quad (24)$$

This matches Puseful,avg already computed in Section 3.4 exactly, because both use the same ṁ, ve, and burn-duration bracket; this is a consistency self-check, not a second independent energy model, and confirms that the existing ve/ṁ/Puseful triple (Sections 3.1-3.4) is internally self-consistent on its own terms. It does not resolve the Eq. 20-23 discrepancy, which concerns whether the Section 3.6 electromagnetic parameters can actually produce that ve/ṁ pair from first principles.

3.8.5 Thrust efficiency

The propulsive/thrust efficiency of the AF-MPD stage specifically is:

$$\eta_{thrust} = P_{kinetic} / P_{input} \quad (25)$$

where Pinput here must mean only the electrical power delivered to the AF-MPD current-drive and field circuit — the portion of the Section 3.4 loss chain labeled "electromagnetic acceleration losses" and "nozzle/exhaust losses" (Section 3.4, items v-vi) — not the total fusion power P_fusion, and not the overall ηtotal defined in Section 3.4. ηtotal already spans the entire chain from fusion yield to useful exhaust kinetic energy (heating/vaporization/ionization, burn fraction, neutron loss, radiation loss, EM coupling, and nozzle losses combined); ηthrust as defined here would be only the EM-coupling/nozzle sub-portion of that chain, isolated from the upstream fusion-related losses. Computing ηthrust against ηtotal would double-count losses already inside ηtotal. This paper does not separately quantify a stand-alone AF-MPD electrical input power distinct from the aggregate ηtotal treatment in Section 3.4:

[PARAMETER NOT ESTABLISHED — REQUIRES SOURCE OR SIMULATION: AF-MPD-stage electrical input power Pinput, and therefore ηthrust, cannot be numerically evaluated with the information available in this paper.]

3.8.6 Parameter sensitivity

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For the simplified model at fixed θ = 90°, thrust scales linearly with each of current, field, and effective length, and with the coupling factor:

$$F \propto \eta F \cdot I \cdot B \cdot L \quad (26)$$

i.e., $\partial F/\partial I = \eta F\ L\ B$, $\partial F/\partial B = \eta F\ I\ L$, $\partial F/\partial L = \eta F\ I\ B$, and $\partial F/\partial \eta F = I\ L\ B$: doubling any single factor doubles F, holding the others fixed, and F is independent of $\dot{m}$ (which instead sets ve = F/$\dot{m}$ for a given F). This is a scaling relationship only, not evidence that thrust can be increased arbitrarily by increasing I or B. Practical ceilings identified elsewhere in this paper constrain all four factors simultaneously: electrode current loading and ablation-driven erosion limits on I and current density (Section 2.3); magnetic-pressure structural stress, found in Section 3.6 to already exceed candidate electrode material strength by 60-200× at the Section 3.6 field level ($\sigma B \approx 64$ GPa); thermal loading from neutron and (partially deposited) alpha flux (Section 3.5); and the pulsed-power driver energy and repetition-rate constraints identified in Section 3.4. Any parameter combination proposed to close the Eq. 20-23 discrepancy must be checked against these limits, not chosen in isolation.

3.8.7 Consistency table

Table 2. AF-MPD parameter consistency check.

Parameter	Existing value	Required/derived	Status
Exhaust velocity ve	50 km/s (3.1)	Pred.(ηF=1): 2.67×10^{8}-2.40×10^{10} m/s	INCONSISTENT (>c)
Mass flow $\dot{m}$	[NOT STATED]	0.752-8.36×10⁻³ kg/s (derived, 3.4 bracket)	DERIVED, not sourced
Current I	10.0 MA (3.6, pinch)	Not est. for AF-MPD stage	AMBIGUOUS
Field B	401 T (3.6, pinch)	Not est. for AF-MPD stage	AMBIGUOUS
Cond. length L	[NOT STATED]	0.05 m (borrowed, 3.5/3.6)	ASSUMPTION
Thrust F	[NOT STATED]	37.6 kN-418 N (ve,$\dot{m}$); 2.01×10^{8} N (Lorentz)	INCONSISTENT (4-6 OoM)
Kinetic power Pk	= Sec.3.4 Puseful	9.40×10^{8}-1.04×10^{7} W	CONSISTENT
Input power P_in	[NOT STATED]	cannot be derived	NEEDS SOURCE/SIM
Thrust eff. ηthrust	[NOT STATED]	cannot compute w/o P_in	NEEDS SOURCE/SIM
Req. ηF (match ve)	-	1.87×10^{-4} (1d); 2.08×10^{-6} (90d)	IMPLAUSIBLY LOW

3.8.8 Scientific status of this section

(A) Derived directly from the proposed equations in this section: Fideal (Eq. 20), ve,predicted (Eq. 21), Fexisting and ηF,required (Eq. 22-23), Pkinetic (Eq. 24), and the $F \propto \eta F\ I\ B\ L$ scaling (Eq. 26).

(B) Adopted from the existing paper: ve = 50 km/s (Sec. 3.1); mGa = 65,000 kg and the 1-90 day burn-duration bracket (Sec. 3.4); I ≈ 10.0 MA and B ≈ 401 T (Sec. 3.6, pinch-confinement stage, reused here only for lack of a dedicated AF-MPD-stage value); L ≈ 0.05 m (Sec. 3.5/3.6 geometry assumption).

(C) Literature-derived: the lumped $F \approx ILB \sin(\theta)$ approximation as a standard first-order MPD-thruster engineering model [5].

(D) Engineering assumptions made in this section: θ = 90° (perpendicular, force-maximizing geometry); reuse of the Section 3.6 pinch I, B and Section 3.5/3.6 geometry L as proxies for AF-MPD-stage values, explicitly flagged as unvalidated for this purpose.

(E) Requires future simulation or experiment: a dedicated AF-MPD-stage current, field, and conductor-length specification distinct from the Z-pinch confinement parameters; the AF-MPD electrical input power Pinput and resulting ηthrust; and the electromagnetic coupling factor ηF itself. A first-order Lorentz-force calculation of the kind performed here does not validate the complete AF-MPD plasma dynamics; a full validation would require, at minimum, an MHD/plasma simulation resolving current-density distribution, magnetic-field distribution, plasma flow, and electromagnetic coupling in the actual thruster geometry.

Net finding: the AF-MPD thrust model, evaluated with the only current and field values available in this paper (the Section 3.6 Z-pinch confinement parameters), is not consistent with the existing ve = 50 km/s / mass-flow assumptions used elsewhere in this paper; the discrepancy is large (orders of magnitude in force, and a non-physical exhaust-velocity prediction) and is reported here rather than concealed. This does not overturn the paper's existing conclusions, which already treat ve = 50 km/s as a literature-comparison-supported design point rather than a first-principles result (Section 3.1), but it does add a specific, previously unidentified gap — a dedicated AF-MPD-stage electromagnetic specification, independent of the fusion-confinement parameters — to the follow-on work already recommended in Section 9.

4. Engineering Constraints and Design Limits

4.1 Capillary Pore Sizing (Young–Laplace Analysis)

The porous electrode structure must retain liquid gallium against surface tension under passive (unpumped) conditions while still permitting capillary-driven flow to the ablating surface during operation. The Young–Laplace relation gives the pressure differential ΔP sustainable across a curved liquid interface of radius r:

$$\Delta P = 2\gamma / r \quad (4)$$

Using the experimentally measured surface tension of liquid gallium at its melting point, γ = 0.708 N/m (verified against sessile-drop measurements reported in the materials-science literature [7]), and a target hold pressure of $\Delta P \approx 1$ atm (1×10^{5} Pa) as a conservative design margin against zero-gravity leakage, the maximum permissible pore radius is:

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$$r = 2\gamma / \Delta P = 2(0.708) / (1\times10^{5}) \approx 1.42\times10^{-5}\ \text{m} \approx 14\ \mu\text{m} \quad (5)$$

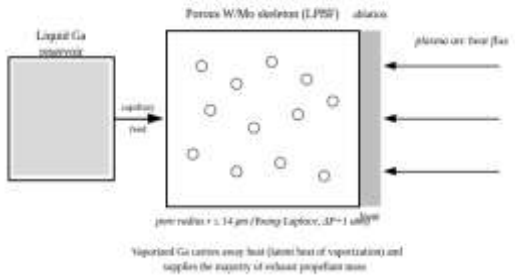


Fig. 5. CPS electrode cross-section: capillary gallium feed through the porous refractory skeleton to the plasma-facing ablation front.

Standard Laser Powder Bed Fusion processes typically achieve minimum feature and pore resolutions on the order of 20–30 μm [8]. The required pore radius therefore sits at or below the practical resolution limit of current LPBF manufacturing — a substantive, currently unresolved constraint on the concept's manufacturability. Closing this gap would likely require either (i) a dedicated post-processing step (e.g., partial sintering, infiltration, or chemical pore refinement) to reduce as-printed pore size below the as-built LPBF resolution, or (ii) relaxation of the passive-retention design requirement in favor of active propellant management (e.g., pressurized or pump-fed gallium delivery), trading the design's passive-retention simplicity for additional moving parts and failure modes.

4.2 Neutron Transport and Electrode Degradation

D-T fusion neutrons are emitted at 14.1 MeV. Because gallium has a comparatively low macroscopic neutron cross-section at this energy, only a small fraction of the 14.1 MeV neutron flux is expected to be absorbed within a sweating layer on the order of 100–500 μm thick; the layer is expected to meaningfully attenuate the alpha-particle and soft X-ray heat load associated with the pinch, but not the fast-neutron flux.

An order-of-magnitude estimate of the neutron mean free path in liquid gallium can be obtained from $\lambda = 1/(n\sigma)$, where n is the atomic number density of liquid gallium ($n \approx \rho N_A/M \approx 5.3\times10^{28}$ m^{-3}, using the measured liquid density $\rho \approx 6.10$ g/cm^3 near the melting point [9]) and σ is the total microscopic cross-section at 14 MeV.

[Flagged placeholder: a total neutron cross-section of σ ≈ 2 barns was used for an illustrative mean-free-path estimate (λ ≈ 9–10 cm); partial (n,2n), (n,p), and (n,n'α) cross-section data for gallium near 14 MeV are reported in the nuclear-data literature [10], but a verified total cross-section could not be confirmed against a nuclear data library (e.g., ENDF/B-VIII.0 or JENDL-5) within the scope of this paper — Insert Data Here: verified total neutron cross-section for natural gallium at 14.1 MeV. The resulting mean free path should be treated as an order-of-magnitude estimate only.]

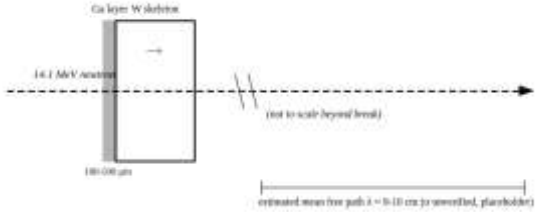


Fig. 6. Neutron mean-free-path schematic (not to scale): the thin Ga layer attenuates thermal/charged-particle flux but not the fast-neutron flux reaching the W skeleton.

Even at this order-of-magnitude level, the estimated mean free path (several centimeters) is one to two orders of magnitude larger than the sweating-layer thickness, indicating qualitatively that the gallium layer's protective role is limited to thermal and charged-particle shielding rather than neutron shielding. The practical consequence is that the underlying tungsten (or molybdenum) skeleton is expected to remain exposed to the full 14.1 MeV neutron flux over the engine's operating life. Fast-neutron irradiation of tungsten is documented in the fusion first-wall materials literature to drive transmutation, principally to rhenium and osmium via neutron capture and subsequent beta decay, along with associated radiation-induced embrittlement and microstructural degradation [11]. Over a multi-burn mission profile, this transmutation and embrittlement process represents a credible electrode lifetime constraint that is not addressed by the gallium ablation-cooling mechanism and should be assessed independently, ideally through a dedicated neutron-transport and materials-damage simulation using verified cross-section libraries.

4.3 Gallium Oxide Passivation

Liquid gallium forms a rigid, sub-nanometer-to-few-nanometer Ga2O3 oxide skin within milliseconds of exposure to trace oxygen or moisture, a well-documented property exploited elsewhere in gallium-based liquid-metal engineering (e.g., shape-retaining soft-robotic and reconfigurable-electronics applications) [12]. In the context of a capillary-fed electrode, this same passivation mechanism poses a risk of pore choking, since residual outgassing of spacecraft materials (H2O, CO2) and any atomic oxygen entrained from prior operation in low Earth orbit could progressively seal the capillary network.

The concept notes propose two mitigations: alloying the gallium with active wetting agents (e.g., titanium or magnesium) to disrupt oxide-layer formation, or periodically flushing the capillary network with hydrogen plasma to reduce the oxide back to metallic gallium.

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[Flagged placeholder: neither anti-passivation mitigation has been demonstrated for this specific capillary-electrode application in the sources reviewed for this paper — Insert Data Here: experimental validation of anti-passivation strategy.]

5. Operational Risk Factors

5.1 Plume Backflow and Spacecraft Contamination

All plasma thrusters exhibit some degree of plume backflow, in which a small fraction of exhaust particles diverge from the primary thrust axis and travel back toward the spacecraft body due to magnetic field divergence and charge-exchange collisions. For gaseous propellants, backflow particles are typically volatile and do not accumulate. Gallium, by contrast, condenses readily on contact with any surface below its 2,204°C boiling point — essentially any spacecraft surface. Backflow-driven metallic film accumulation on solar arrays, optical sensors, thermal radiators, and exposed electronics is therefore a credible contamination risk specific to a condensable liquid-metal propellant, and would need to be bounded through plume-divergence modeling and, likely, physical baffling or standoff geometry between the thruster and sensitive spacecraft surfaces before flight qualification.

5.2 Tritium Fuel-Cycle Constraint

Tritium is not naturally abundant and decays with a 12.3-year half-life, precluding long-term stockpiling. Civilian tritium supply is dominated by Canada's fleet of CANDU heavy-water reactors, which collectively produce on the order of 2 kg of tritium per year as a byproduct of reactor operation, against a global civilian stockpile estimated at approximately 25–30 kg; these figures are broadly consistent across multiple recent policy and technical sources reviewed for this paper [13].

Under the (unverified, lower-bound) 0.24 kg D-T fuel-mass estimate developed in Section 3.3, and assuming a D-T mixture split evenly by mole between deuterium and tritium, a single engine burn could require on the order of a tenth of a kilogram of tritium — a non-trivial fraction of annual global civilian tritium production. This demand would compete directly with the tritium requirements of terrestrial fusion research programs, including ITER, and would require a dedicated, likely government-controlled, fuel-procurement and handling pathway rather than commercial off-the-shelf sourcing.

5.3 Liquid-Metal Slosh and Attitude Control Interaction

A propellant load on the order of tens of tonnes of liquid gallium (density ≈ 6.1 g/cm³, roughly six times that of water) represents a substantial, mobile mass fraction of the vehicle. In microgravity, an unbaffled liquid mass of this scale is susceptible to slosh dynamics that can couple destructively with the spacecraft's attitude control system during maneuvers, particularly given gallium's high density relative to conventional cryogenic propellants. This is a well-characterized class of problem in spacecraft propellant-tank design, typically addressed through slosh baffles, compartmentalized tankage, or slosh-mass dynamic models integrated into the attitude control system design. The concept notes do not yet specify a tank or baffle architecture; a coupled slosh–attitude-control analysis is recommended as a design task rather than treated as resolved.

6. Regulatory and Planetary Protection Considerations

6.1 Treaty Framework for Pulsed Nuclear Propulsion

The 1963 Partial (Limited) Test Ban Treaty prohibits nuclear weapon test explosions in the atmosphere, in outer space, and underwater, while the 1967 Outer Space Treaty's Article IV prohibits placing nuclear weapons or other weapons of mass destruction in orbit or on celestial bodies. Neither treaty was drafted with pulsed fusion propulsion specifically in mind, and whether a device that produces thrust through repeated small-scale fusion pulses would be classified as a prohibited nuclear test or weapon, as opposed to a propulsion system, is not settled by the treaty text itself [14,15].

This ambiguity is not new: it was a significant factor, alongside the 1963 treaty's atmospheric and space testing provisions, in the termination of Project Orion, the 1950s–1960s nuclear-pulse propulsion program [14], and it remains an open legal question for any pulsed nuclear propulsion concept, including the one considered here. Resolving it would require a dedicated legal and policy analysis beyond the scope of this engineering paper; the concept should not be assumed to be legally unproblematic on the basis of the physics analysis alone.

6.2 Planetary Protection

Neutron activation of the gallium propellant during the fusion pulse could render the exhaust plume mildly radioactive, depending on the specific activation products and half-lives involved, which are not quantified in this paper. COSPAR planetary protection policy restricts the introduction of radioactive or biological contamination to solar-system bodies of interest for prebiotic chemistry or the search for life, including Mars [16]. A mission profile that involves directing an activated exhaust plume toward, or into the upper atmosphere of, such a target body would need to demonstrate compliance with these restrictions, likely through activation-product inventory modeling and plume-dispersion analysis, before mission approval could be expected.

6.3 Launch Authorization

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In the United States, the launch of any spacecraft containing a space nuclear system is governed by National Security Presidential Memorandum 20 (NSPM-20, issued August 20, 2019), which established a risk-informed safety analysis and launch-authorization process administered through the Interagency Nuclear Safety Review Board [17]. Comparable systems (radioisotope power sources, in past NASA missions) have required safety-review processes spanning several years and substantial dedicated budget, independent of the underlying spacecraft's own development schedule. A pulsed-fusion propulsion system carrying tritium and generating a neutron flux during operation should be expected to fall under an equivalent, and likely more extensive, review process; this represents a major programmatic and schedule driver that should be planned for from an early design phase rather than treated as a late-stage formality.

7. Candidate Mission Profiles

The milli-g-class thrust levels reported for comparable published magneto-inertial fusion propulsion concepts (Section 3.1) imply that this class of engine is best suited to missions that can tolerate a sustained, low-thrust acceleration profile rather than a short impulsive burn. Combined with the unresolved contamination, neutron-shielding, and regulatory questions identified in Sections 4–6, the most plausible near-to-mid-term applications are uncrewed rather than crewed:

- High-Δv outer-planet orbiter delivery (e.g., a Uranus or Neptune orbiter). Sustained high-Isp thrust could reduce trip time and propellant mass relative to the chemical-propulsion-plus-gravity-assist trajectories used by historical outer-planet missions, at the cost of requiring low-thrust trajectory optimization rather than impulsive maneuver planning.
- Uncrewed bulk cargo pre-positioning in support of a crewed Mars campaign architecture. Routing high-Δv, radiation- and contamination-sensitive propulsion technology through cargo-only vehicles decouples it from the human-safety-critical path while still reducing the overall chemical-propellant mass required for a crewed campaign.
- A fast outer-solar-system or heliopause precursor probe, benefiting from sustained low-thrust acceleration over a multi-year cruise phase, in the spirit of prior electric-propulsion precursor mission concepts but with a potentially higher achievable cumulative Δv.

Mission Profile	Regime	Key Driver	Crew Status
Outer-planet orbiter delivery (e.g. Uranus/Neptune)	Sustained low-thrust, multi-year cruise	High Isp reduces trip time / propellant vs. chemical-gravity-assist	Uncrewed
Bulk cargo pre-positioning	Low-thrust, non-time-critical	Decouples novel [illegible] from crew-critical path	Uncrewed
Heliopause / outer-system precursor	Continuous thrust, multi-year cruise	High cumulative Δv over long cruise	Uncrewed

Fig. 7. Candidate mission profiles compared by thrust regime, key performance driver, and crew status.

Crewed applications are deferred pending resolution of the contamination, neutron-shielding-mass, tritium-supply, and regulatory questions identified in Sections 4–6; none of these are dismissed as unsolvable, but none are yet resolved to a level that would support a credible crewed mission design.

8. Supporting Infrastructure Requirements

Maturing this concept toward flight readiness would require ground-based infrastructure substantially beyond what is needed for a conventional electric or chemical propulsion system:

- A secure, likely government-coordinated tritium supply chain and handling facility, given the scarcity of civilian tritium and its competing use in terrestrial fusion research programs (Section 5.2).
- Specialized Laser Powder Bed Fusion manufacturing and non-destructive evaluation capability qualified for sub-20-μm porous refractory-metal structures, or an equivalent post-processing capability to reach the pore sizes required in Section 4.1.
- Ground-based pulsed-power and plasma test facilities capable of simulating Z-pinch-relevant current and neutron environments, comparable in class to existing national-laboratory pulsed-power facilities used for Z-pinch and inertial confinement fusion research.
- Radiation-hardened power management and avionics for the pulsed, high-current drive electronics, and for any onboard systems located near the neutron-emitting engine.
- On-orbit assembly and fueling infrastructure, given the large absolute propellant tank volumes implied by a multi-tonne gallium load, even accounting for gallium's favorable density relative to cryogenic propellants.
- A nuclear launch safety review and authorization pathway (e.g., the NSPM-20 / Interagency Nuclear Safety Review Board process in the U.S. context, or an equivalent framework in other launching states), understood as a multi-year, resource-intensive programmatic requirement rather than a technical footnote (Section 6.3).

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- A planetary protection compliance review, particularly for any mission profile that would direct an exhaust plume toward a body of astrobiological interest (Section 6.2).

9. Conclusion

This paper has formalized the operating principles of a gallium-fueled fusion-choke AF-MPD thruster concept and evaluated its performance envelope against standard propulsion and plasma-physics relations. The concept's assumed exhaust velocity range is broadly consistent with values reported for comparable published magneto-inertial fusion propulsion studies, and the associated propellant mass-fraction reduction relative to chemical propulsion is substantial in an idealized, impulsive-Δv comparison.

At the same time, this analysis has identified several specific points in the underlying concept notes that require further work before the concept's feasibility can be credibly assessed: an unstated fusion-to-propellant energy coupling efficiency that renders the fuel-mass estimate a lower bound rather than a design figure; an unreconciled discrepancy between two independently assumed propellant-mass scenarios; an unverified neutron cross-section value underlying the electrode neutron-shielding assessment; a capillary pore-size requirement at or below current additive-manufacturing resolution; and open questions regarding spacecraft contamination, tritium supply, liquid-metal slosh dynamics, and treaty-level regulatory status.

Recommended follow-on work includes: (i) energy-balance modeling using a realistically bounded fusion-to-propellant coupling efficiency rather than an idealized 100% conversion assumption; (ii) verification of the gallium neutron cross-section against a formal nuclear data library; (iii) reconciliation of the vehicle propellant-mass budget across a single, internally consistent set of assumptions; and (iv) replacement of the impulsive rocket-equation estimate with a low-thrust trajectory optimization appropriate to the concept's milli-g-class acceleration regime. None of the constraints identified here appear to be fundamental physical impossibilities; each instead defines a specific, tractable area of further engineering and materials-science study.

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